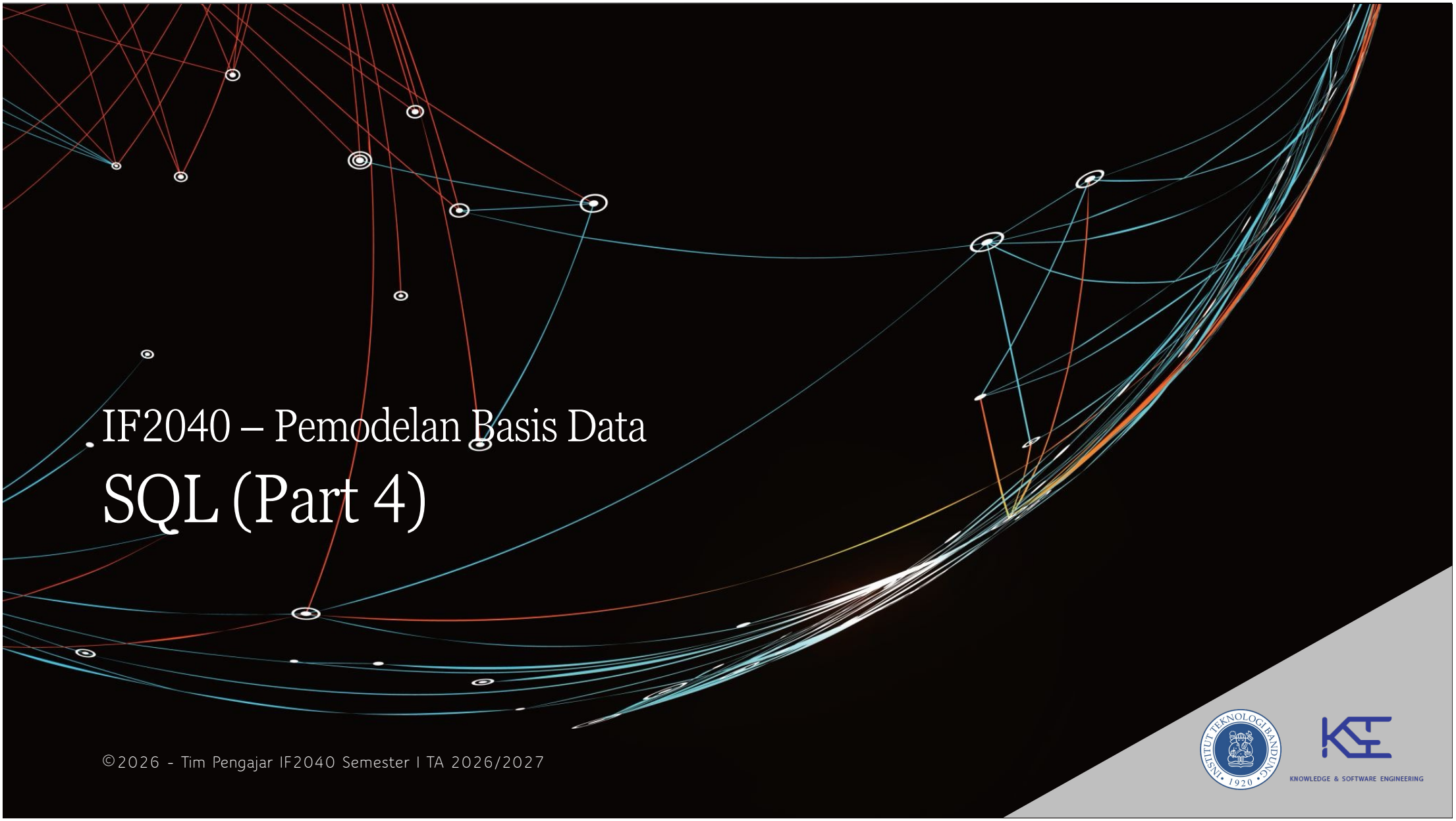
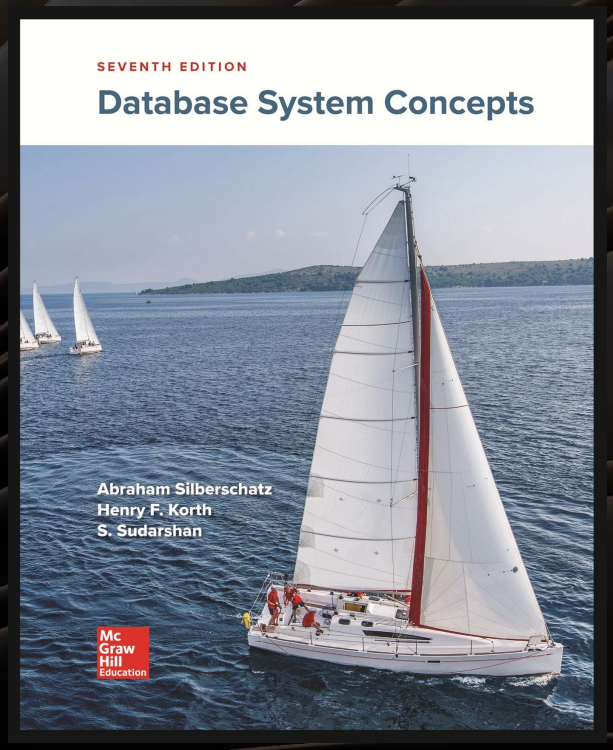


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SQL (Part 4)

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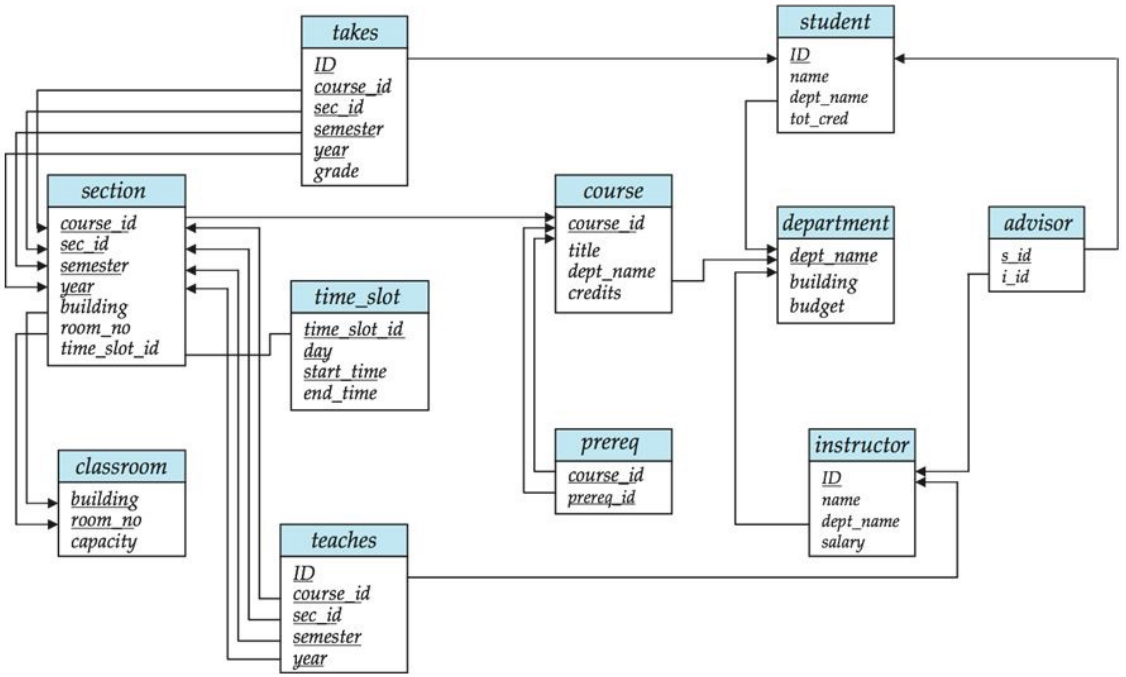
Sumber

Silberschatz, Korth, Sudarshan: "Database System Concepts", 7th Edition

- Chapter 3 : Introduction to SQL
- Chapter 4 : Intermediate SQL

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Schema Diagram



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Data Definition Language

Data Definition Language

The SQL data-definition language (DDL) allows the specification of information about relations, including:

1. The schema for each relation.
2. The type of values associated with each attribute.
3. The Integrity constraints
4. The set of indices to be maintained for each relation.
5. Security and authorization information for each relation.
6. The physical storage structure of each relation on disk.

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Domain Types in SQL (1)

char(n). Fixed length character string, with user-specified length *n*.

varchar(n). Variable length character strings, with user-specified maximum length *n*.

int. Integer (a finite subset of the integers that is machine-dependent).

smallint. Small integer (a machine-dependent subset of the integer domain type).

numeric(p,d). Fixed point number, with user-specified precision of *p* digits, with *d* digits to the right of decimal point. (ex., **numeric**(3,1), allows 44.5 to be stores exactly, but not 444.5 or 0.32)

real, double precision. Floating point and double-precision floating point numbers, with machine-dependent precision.

float(n). Floating point number, with user-specified precision of at least *n* digits.

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Domain Types in SQL (2)

date Dates, containing a (4 digit) year, month and date
- Example: **date** '2005-7-27'

time Time of day, in hours, minutes and seconds.
- Example: **time** '09:00:30' **time** '09:00:30.75'

timestamp date plus time of day
- Example: **timestamp** '2005-7-27 09:00:30.75'

interval period of time
- Example: interval '1' day
- Subtracting a date/time/timestamp value from another gives an interval value
- Interval values can be added to date/time/timestamp values

Each type may include a special value called the null value.
- A null value indicates an absent value that may exist but be unknown or that may not exist at all.

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Create Table Construct

An SQL relation is defined using the **create table** command:

create table *r*

(A_1 D_1, A_2 D_2, ..., A_n D_n,
(integrity-constraint$_1$),
...,
(integrity-constraint$_k$))

- *r* is the name of the relation
- each A_i is an attribute name in the schema of relation *r*
- D_i is the data type of values in the domain of attribute A_i

Example:

create table *instructor* (

ID	**char**(5),
name	**varchar**(20),
dept_name	**varchar**(20),
salary	**numeric**(8,2))

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Integrity Constraints in Create Table

Types of integrity constraints

- **primary key** (A_1, ..., A_n)
- **foreign key** (A_m, ..., A_n) **references** *r*
- **not null**

SQL prevents any update to the database that violates an integrity constraint.

Example:

```
create table instructor (
        ID              char(5),
        name            varchar(20) not null,
        dept_name       varchar(20),
        salary          numeric(8,2),
           primary key (ID),
           foreign key (dept_name) references department);
```


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And a Few More Relation Definitions

```
create table student (
        ID                  varchar(5),
        name                varchar(20) not null,
        dept_name           varchar(20),
        tot_cred            numeric(3,0),
           primary key (ID),
           foreign key (dept_name) references department);

create table takes (
        ID                  varchar(5),
        course_id           varchar(8),
        sec_id              varchar(8),
        semester            varchar(6),
        year                numeric(4,0),
        grade               varchar(2),
            primary key (ID, course_id, sec_id, semester, year) ,
            foreign key (ID) references  student,
            foreign key (course_id, sec_id, semester, year) references section);
```

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And more still

```
create table course (
        course_id       varchar(8),
        title             varchar(50),
        dept_name       varchar(20),
        credits           numeric(2,0),
            primary key (course_id),
            foreign key (dept_name) references department);
```


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Not Null Constraints

not null

- Declare *name* and *budget* to be **not null**

```
name varchar(20) not null
budget numeric(12,2) not null
```


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Default Constraints

is used to set a default value for a column

Declare default value for budget is 500.000

```
name varchar(20) not null
budget numeric(12,2) default 5000000
```





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Unique Constraints

unique (A_1, A_2, …, A_m)

- The unique specification states that the attributes A_1, A_2, …, A_m form a candidate key.
- Candidate keys are permitted to be null (in contrast to primary keys).

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Referential Integrity

Ensures that a value that appears in one relation for a given set of attributes also appears for a certain set of attributes in another relation.

- Example: If "Biology" is a department name appearing in one of the tuples in the *instructor* relation, then there exists a tuple in the *department* relation for "Biology".

Let A be a set of attributes. Let R and S be two relations that contain attributes A and where A is the primary key of S. A is said to be a foreign key of R if for any values of A appearing in R these values also appear in S.

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Referential Integrity (Cont.)

Foreign *keys can be* specified as part of the SQL **create table** statement

foreign key (*dept_name*) **references** *department*

By default, a foreign key references the primary-key attributes of the referenced table.

SQL allows a list of attributes of the referenced relation to be specified explicitly.

foreign key (*dept_name*) **references** *department* (*dept_name*)

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Cascading Actions in Referential Integrity

When a referential-integrity constraint is violated, the normal procedure is to reject the action that caused the violation.

An alternative, in case of delete or update is to cascade

```
create table course (
   (…
    dept_name varchar(20),
    foreign key (dept_name) references department
         on delete cascade
         on update cascade,
      . . .)
```

Instead of cascade we can use :

- set null,
- set default

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Updates to tables

Insert

- insert into *instructor* values ('10211', 'Smith', 'Biology', 66000);

Delete

- Remove all tuples from the *student* relation
  - delete from *student*

Drop Table

- drop table *r*

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Updates to tables

Alter

- alter table *r* add *A D*
  - where *A* is the name of the attribute to be added to relation *r* and *D* is the domain of *A*.
  - All exiting tuples in the relation are assigned null as the value for the new attribute.
- alter table *r* drop column *A*
  - where *A* is the name of an attribute of relation *r*
  - Dropping of attributes not supported by many databases.
- alter table *r* rename column *old_name* to *new_name*
  - to rename a column in relation *r*.
- alter table *r* alter column *A D*
  - to change the datatype of column *A* in relation *r* to *D*.

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